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Understanding the True Potential of Complex Turbidite Reservoirs

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Abstract

Technological advances in tools and processing techniques are playing a crucial role in evaluating discoveries of the past. This paper offers a unique workflow to assess the potential of complex turbidite reservoirs which would have otherwise been overlooked using conventional wireline logging tools and traditional petrophysical techniques.

In highly heterogenous reservoirs below wireline logging tool resolution, conventional interpretation techniques do not provide an accurate representation of reservoir quality. The primary focus of this paper is to demonstrate how a robust understanding can be established on fine-scale variations in reservoir quality of complex turbidite sandstones by integrating geological and advanced petrophysical techniques. The proposed workflow integrates a neural net approach by utilizing data from cores, petrography, borehole images, and NMR log.

Petrography and MICP data were utilized to understand diagenetic control on pore-throat radii and reservoir quality change due to grain properties (size, sorting, maturity), dissolution, and calcite cementation. Core calibrated image logs were used to classify rock types and train the neural net. The outcome is a high-resolution core calibrated rock-type model which has played a critical role in distinguishing sweet spots from poor quality reservoir intervals in these complex turbidites. The model has proven over 80% accurate when blind tested against cores and dynamic data in offset wells. This has subsequently helped in generating high confidence net to gross estimations across the field to substantiate the potential in these turbidite units.

This novel methodology has shown extremely promising results in understanding the degree of heterogeneity in complex turbidites. The new technique has allowed identification, location, and scaling of by-passed zones which were overlooked previously using low resolution logging tools and traditional petrophysical processes.

Introduction

Petrophysical detection of highly laminated sand-shale deposits has always been a challenge in the hydrocarbon industry (Worthington, 1997; Hansen & Fett, 2000; Passey et al., 2006; Bastia et al., 2007; Joubert & Maïtan, 2010; Yadav et al., 2012). Such reservoirs are being consistently overlooked due to the lack of vertical resolution in conventional logging tools. The reserves equation requires net-sand to gross-rock (NTG) as an input. However, as net-sand is usually calculated through applying cutoffs on conventional logging tools, the results are an approximation, and not true representations of the net reservoir thickness. This is an industry wide problem that continues to exist until today. Hansen and Fett (2000) utilized high-resolution logs instead of low-resolution ones, where they carefully integrated the micro-resistivity image-log to conventional cores and side-wall cores to define a set of electrofacies.

Thin beds are a geological phenomenon that is typically associated with certain depositional environments. The first environment is of a tidal-flat reservoir, where the sands are so thin that ~ 80 % of hydrocarbon is stored within 2.5 cm thick layers (Passey et al., 2006). The second environment is of deep marine turbidites reservoirs, where ~50% of the hydrocarbon is stored in mere 2.5cm thick layers (Worthington, 1997; Hansen & Fett, 2000; Passey et al., 2006; Joubert & Maïtan, 2010).

This paper provides petrophysical evaluation of the heterogeneous thin-bedded sands of a reservoir by providing an advanced thin bed analysis workflow that integrates geological understanding, core descriptions, special core analysis (SCAL) measurements and artificial neural network (ANN) model. The reservoir in study is composed of deep marine turbidites, depositing conglomerates and massive poorly consolidated sandstones. The upper part of the reservoir is comprised of calcareous mudstones. Laminated sand-shale sequences also exist as a result of reoccurring submarine fan deposition. The aim is to address the inherent issue in thin bed evaluation of reservoirs due to unaccounted for 'net reservoir'.

Methodology

To understand the underlying controls on reservoir heterogeneity, laboratory tests were performed on core samples including porosity and permeability measurements, thin-section analysis, X-ray diffraction (XRD) for mineral composition and quantification, and mercury injection capillary pressure (MICP) to study the pore throat size distribution. Petrophysical analysis and core-to-log integration was performed using technical log evaluation software. The methodology can be broken down into three main parts following the objectives of this study (Figure 1).

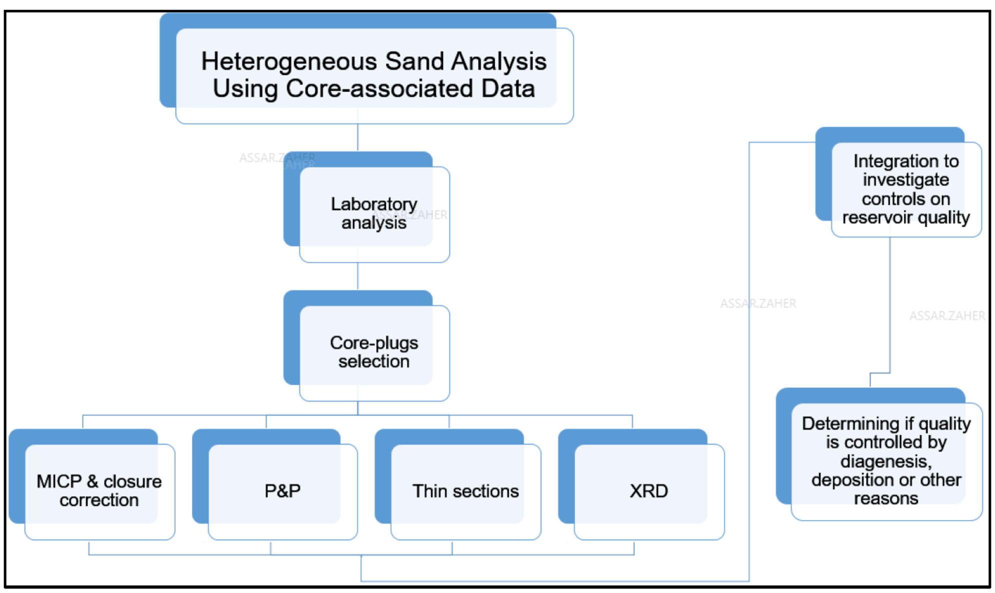

Figure 1—A summary of the heterogeneous sand analysis using core-associated data in this study.

- Heterogeneous sand analysis with MICP: Integrating thin-sections with MICP data to understand the underlying cause for the reservoir sands heterogeneity.

- Handling the heterogeneity (rock-type modeling): Utilizing high resolution synthetic resistivity log (SRES) extracted from image logs and NMR, calibrated by core data, to create a high-resolution understanding of the heterogeneous reservoir.
- Artificial neural networks analysis: Employing machine learning to populate the high-resolution rock typing workflow in nearby wells.

Since conventional logs do not capture the heterogeneity of the complex sandstone reservoir, a higher-resolution solution should be applied. Building upon the work of Hansen and Fett (2000), a similar approach is used in this study. The approach involves core images, reservoir-representative thin-section images, porosity and permeabilities from core plugs, NMR porosity log, static and dynamic image logs, and the SRES log (Figure 2). SRES log is a synthetic resistivity log that is extracted from the image-log resistivity measurements. The SRES log is the average of all image-log sensors readings at 0.1"-0.2", thus providing a very high vertical-resolution measurement.

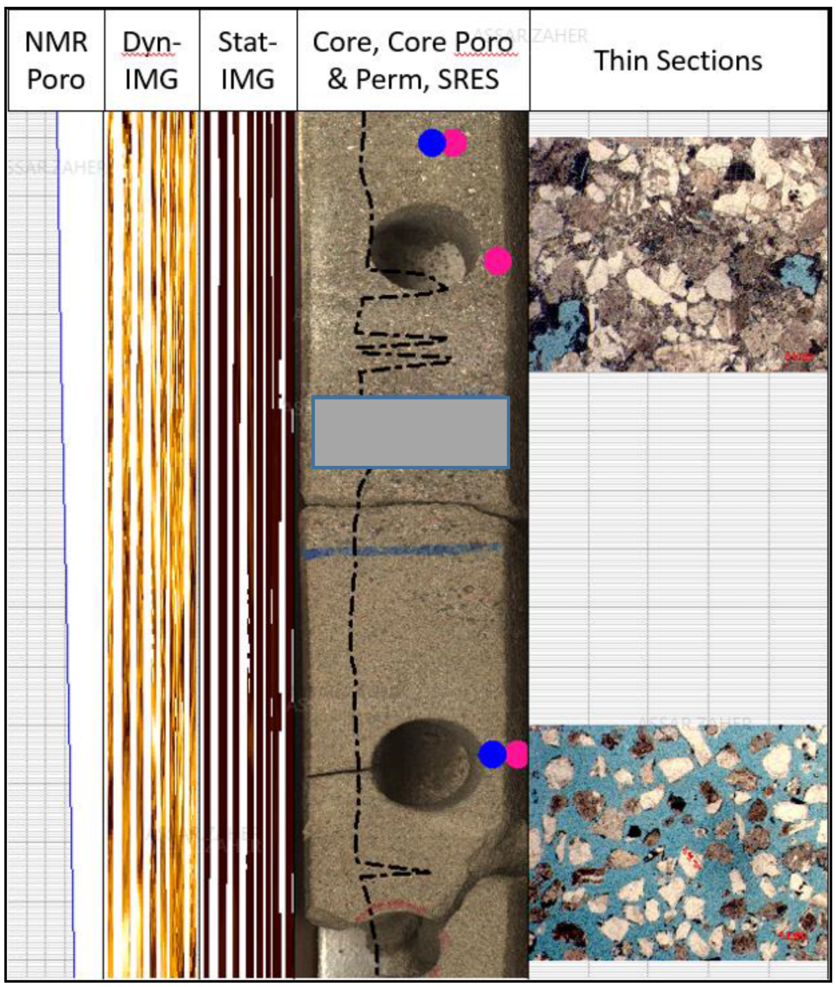

Figure 2—The foundation of the rock-type model: A display that integrates core-associated data with high resolution petrophysical logs. This display is used to define SRES windows based on reservoir quality of rock-types.

The modeling method consists of distinguishing between each type of facies present within the core and core-data, and classifying them based on their permeability. Five main facies were defined based on permeability cutoffs: Good-sand, Shaly-sand, Tight-sand, Shale and Non-reservoir. SRES cut-offs were applied across the cored intervals with core porosity and permeability data in order to train the machine learning. A module of the technical log evaluation software was used as a neural network application. This subsequently helped in scanning the entire available core to the resolution of the image log. NMR porosity is used to help the image-log distinguish between reservoir sand and non-reservoir sand. By applying the NMR-porosity cutoff and the SRES cutoff to the image-log data set, a rock-type model was generated (Figure 3).

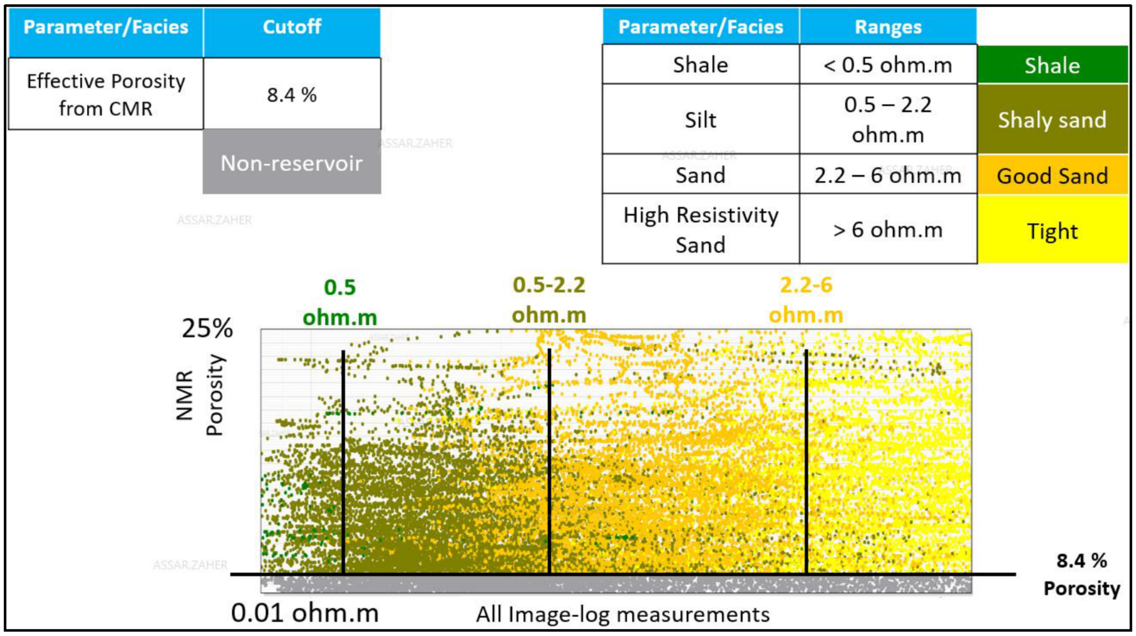

Figure 3—Cutoffs applied on SRES log and NMR porosity to establish a rock-type model that accurately captures good quality reservoir sands based on calibrations with core data. Lithology with NMR porosity below 8.4% is considered non-reservoir by the model. Lithology with image-log measurement of <0.5 ohm.m is counted as shale, 0.5-2.2 ohm.m as shaly-sand, 2.2-6 ohm.m as good-sand, and >6 ohm.m as tight-rock.

The following logs were used as input data to help the evaluation software predict the rock-type model.

1. High-resolution density
2. Photoelectric effect
3. High-resolution compressional slowness log
4. Potassium and thorium concentrations
5. SRES

Once the logs are loaded, the neural network runs a "supervised test". The machine uses the provided logs to produce a predicted facies curve similar to the set reference, which is the rock-type model (Figure 4). As a default, all logs are set to have the same weighting in deciding what different facies the prediction produces. In clean homogeneous reservoirs, logs having the same weighting can lead to good correlations between reference and prediction. However, in complex heterogeneous reservoirs, changing the weighting of logs can help achieve higher correlations. Some logs create better predictions for different reasons, for example, due to their vertical resolution, or their sensitivity to present mineralogy.

Since the reservoir under study is complex and heterogeneous, an alternative method is required to achieve high correlation between prediction and reference. The method involves setting individual logs to make a prediction alone. The same process is repeated with every log, and separate predictions are produced. This is done to lower the "black-box" effect of the neural network, and to understand what each log is contributing to the prediction. Once all logs undergo the same process, they are displayed next to each other along with the reference (Figure 4). The weighting of each log is adjusted accordingly to improve the quality of the prediction.

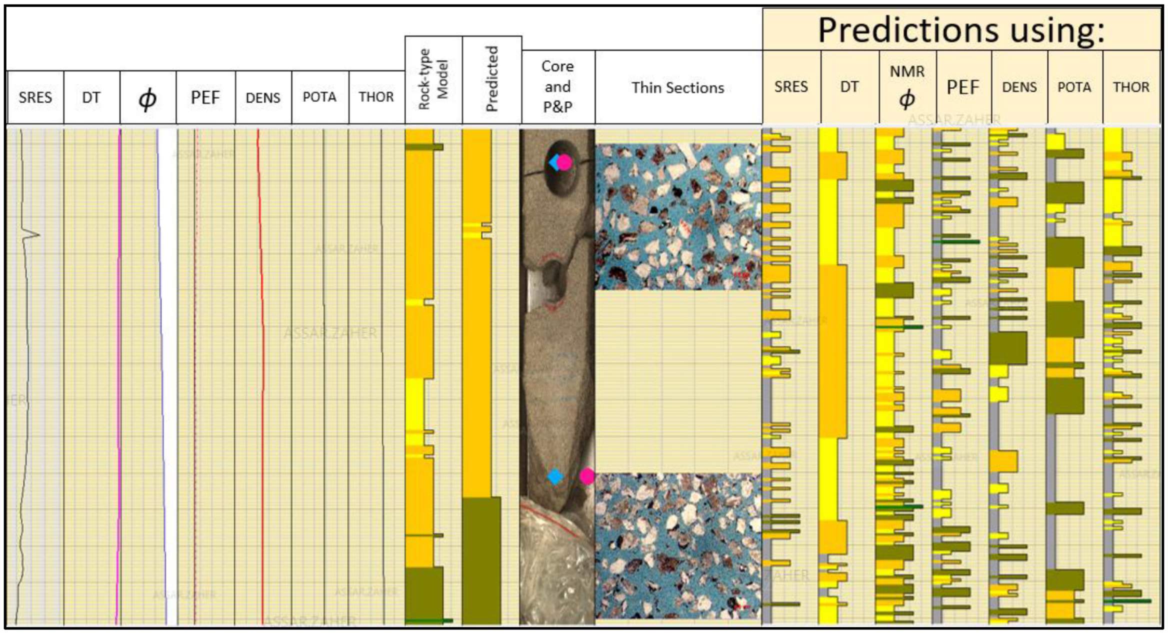

Figure 4—Logs utilized for artificial neural networks plotted against core-data to quality-check the accuracy of predictions. Both thin-sections were taken from a very friable sand interval. The loose sand particles were assembled together using epoxy resin, and are not representative of the reservoir condition.

Rock-type modeling is a process that integrates a variety of data from different sources. This can introduce errors in the results if not understood and thoroughly quality checked. In the following discussion, some of these uncertainties will be addressed in detail along with the way to tackle them where possible.

Porosity and permeability measurements can be acquired from either partially or fully cleaned core plugs. Partial cleaning is quicker, but does not completely remove liquids that reside within the plug. Liquids can reduce the permeability of the sample, especially in plugs with permeability values between 2 and 20 mD, while having almost no effect on samples with 100 mD of permeability or higher (Worthington, 1990). Fully cleaning the core plug removes residing liquids and can produce more accurate permeability measurements. Out of the core plugs used as calibration points for the rock-type model, 10/48 were partially cleaned and the rest was fully cleaned. Such discrepancies in plug preparation make permeability measurements at lower permeability thresholds uncertain. These permeability values are used in the model's calibration, and potentially cause miscorrelation or errors in choosing certain rock-types cutoffs.

Reservoir rock condition differs from the conditions where core plugs are tested at surface conditions. This is particularly true when it comes to compaction, as it can significantly reduce the permeability of sands (Schutjens & de Ruig, 1997; Ostermeier, 1995; McLatchie et al., 1958). The sands analyzed ranged from loose unconsolidated sands, to over-pressurized and well-consolidated sands. Schutjens & de Ruig (1997) showed a decrease in permeability associated with compaction between 15 to 25%. Core-plugs tested for porosity and permeability in this study did not undergo stress or compaction analysis, which could to reduce the accuracy of the permeability values. To mitigate this limitation, a wider permeability cutoff range was selected for different types of facies (< 1 mD for tight, 1-60 mD for shaly-sands, and > 60 mD for good-sands). Core-plug stress analysis should be incorporated in the future as a routine step in the workflow to reduce uncertainty associated with porosity and permeability values.

Lofts & Bristow (1998) promoted depth-shifting the core directly using image-log tools. They displayed 360° digital core images against the image-log and applied a depth-shift using distinct features that appeared on both mediums. Features included brecciated limestones and irregularly dipping fractures. Digital core images were rotated to improve the accuracy and precision of the correlation to the image-log.

On the contrary, the core depth-shift in this study was not directly performed using an image-log. Instead, the core was depth-shifted using core gamma and wireline Gamma Ray Log. This decision was reached after multiple failed attempts to correlate the core directly to the image-log. Attempts failed for two reasons: 1) image-log quality and coverage were low, 2) the core lacked distinctive features that correlate to the image-log. These reasons are discussed further below.

The coring plan for Well-1 was to acquire two consecutive cores of 60' each, reaching a total of 120' of core. Unfortunately, only 110' of core were recovered, meaning there are 10' missing. In the industry, it is always assumed that the missing section is at the bottom of the core, but that is not always the case (Lofts & Bristow, 1998). Two consecutive cores of ~110' were used for the depth-shifting. In some cases, consecutive coring can result in the loss of +/- 1' of core in the process of tripping out and in, especially if the lithology is friable between cores. While one foot might seem insignificant, it can make depth-shifting very difficult when combined with other core and image-log uncertainties.

If the reservoir sands are filled with gas, a degassing effect occurs. The core expands when reaching equilibrium at surface temperature and pressure (Lofts & Bristow, 1998). This expansion leads to matching errors between core images the image-log since one is at surface conditions, while the other is at reservoir conditions.

Lofts & Bristow (1998) pointed to another issue related to depth-shifting using the image-log. Image-logs are subject to many errors associated with wireline acquisition. One of these problems is the "yoyo" effect, where the tool is jammed in the formation momentarily before being forcefully freed by the tension of the winch, resulting in a short period of vertical oscillation from the tool. Wireline tools are designed to be constantly measuring as a function of time, not depth. This causes a mismatch between the actual depth of the tool and the depth being recorded in the surface computer. Many other corrections are applied to the image-log and are discussed in detail by Passey et al (2006). While the corrections are well established and understood, the process of correction still produces uncertainties to the image-log.

Lack of distinctive features on the core was another major reason why direct core to image-log correlation was very uncertain. The 110' of cores were completely formed of interbedded sands and shales, with some thicker sand intervals randomly distributed. No visible calibration points were found to give confidence in depth matching.

Adding even more complexity to the depth-shift process, the image-logging tool utilized in this study was designed for maximum coverage in 8'' wellbore size. However, the reservoir section of Well-1 was drilled in 12'' wellbore size, which reduced the image-log coverage to 65.36% down from 98% usual coverage in 8'' wellbores. The reduced coverage made it even more difficult to make a certain depth-shift. The obvious solution for this issue is to preemptively plan to drill thin-bedded reservoirs in 8'' wellbore size, or to develop the tool further to provide more coverage in larger wellbore sizes. Figure 5 shows a comparison between the quality of two images acquired in two different wells (Well-1 and Well-X) with different wellbore sizes (8'' and 12''). Image quality of 8'' wellbore size is far superior in quality and clarity. SRES seems to respond to small scale lithological changes in 8'' much more accurately than it does in the 12'' wellbore. A depth shift of the 60' of core in Well-X was done using direct core to image-log correlation. The entire core was easily correlated and accounted for on the image-log, foot by foot. This success is not only due to the better image-log quality, but because there are many clear anchor points in terms of lithology like the presence of pebbly sandstones and conglomerates. Another possible solution is to acquire side-wall cores before imaging the borehole. Theoretically, this allows using the circular voids left from the side-wall coring process as a visual method of depth matching the image-log. Direct calibration is possible because the side-wall cores and image-logs are both run on wireline and share the same depth referencing method.

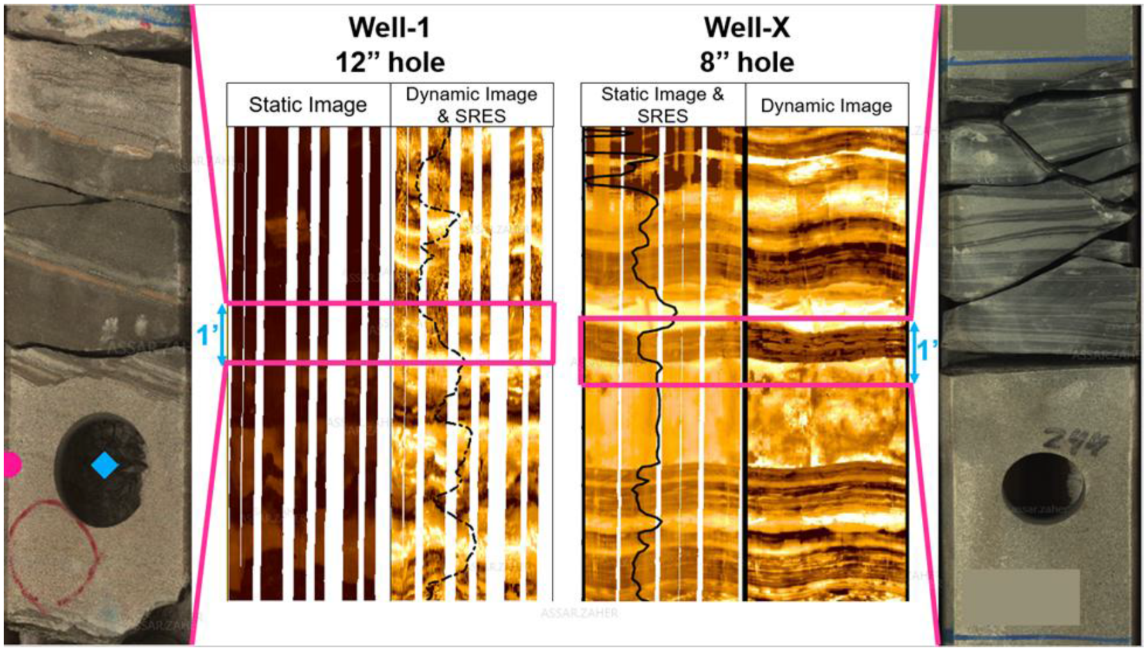

Figure 5—A comparison on the same scale between the image-log quality in Well-1 versus Well-X. Well-1 acquired image-log in a 12″ wellbore, while Well-X acquired it in an 8″ wellbore. The quality of image in the 8″ wellbore is far superior in terms of clarity for lithological interpretation. SRES in 8″ seems to capture small scale lithology changes in a much better way than SRES in the 12″ wellbore.

As defined previously, SRES log does not accurately represent the 360° array of resistivity measurement the image-log tool takes at one depth. Instead, SRES is the synthetic average of the array measurements. Having one value instead of an array is a simplification that helps guide the cutoff process, but it also biases the interpreter. The process entails correlating certain SRES values to certain rock-types. The image-log could be reading 49% of shaly-sand and 51% of good-sand around the wellbore at a certain depth. Such interval should have the interpreter be pessimistic in setting the cutoff for rock-typing. If observations of SRES and associated P&P measurements yield good reservoir properties values, interpreters will be inclined to apply an optimistic cutoff. This case disregards the potential of biased plugging where the rock looks more intact and easier to plug at a given interval.

The sand count module in the software utilized for modeling attempts to avoid such bias by using the image-log array readings as an input for the rock-type model. Figure 6 shows how the module bins the image to extract a fraction of different rock-types from it based on resistivity values. The module then assigns the rock-type with the highest fraction as the dominant rock-type for the model. While this does eliminate some of the bias, it is good for the interpreter to quality check the model against the binned image. However, The SRES log in Figure 2 and Figure 4 appears to be spikey, and these spikes seem to occur randomly throughout the acquired interval. Whenever these spikes occur, it is highly likely that the image-log is unreliable at the locality, and will produce rock-typing errors. This coincides with two points that were of shaly-sand properties (Figure 3) with permeability values of 7 and 20 mD. SRES value at these two points ranged between 10 to 16 ohm.m, exceeding the threshold for shaly-sand and good-sand, and landing well beyond the tight-rock cutoff. Many possible explanations for this phenomenon are feasible. It could occur because there's a depth-shift issue at the locality. It could also occur due to a yoyo effect on the image-log. Another possibility is that image-log malfunctioned at the location or was not corrected in a proper manner, producing spikes that measure much higher than possible in that type of lithology.

Figure 6—This figure shows how the model extracts rock-types from the image-log based on 360 degrees of readings around the well bore. At the same depth, the image-log could measure the resistivity of tight, good sand, and shale. These measurements are shown as fractions, the rock-type with the largest contribution to these fractions is considered as the overall lithology by the rock-type model.

Passey et al (2006) mentioned the usefulness of NMR as a tool to evaluate thin-beds. Minh & Sundararaman's (2006) experiment supports this argument. Though NMR lacks the vertical resolution to resolve individual thin-beds, it can indicate the presence of hydrocarbon within them in ideal conditions. NMR in the case of this study is used as a lithology variance tool and not as a hydrocarbon indicator. To generate a model using the sand count module of the modeling software, a porosity curve is required. A cutoff is applied on the curve to distinguish reservoir from non-reservoir. Since the NMR's vertical resolution is low (~4'), using it negatively impacts the model's quality. NMR porosity is used in this study for lack of better alternatives. This is problematic because the sand-count module in the modeling software prioritizes the porosity cutoff over the image-log cutoff. Meaning if a certain interval had 8.4% porosity, the model will immediately consider it as non-reservoir even if it had the image-log resistivity value of 2.2-6 ohm.m, which meets the good sand cutoff. Setting the cutoff on NMR porosity is an iterative process to find a compromise that matches the most calibration points.

The uncertainty of using a low-resolution log as a porosity indicator is carried within the model. No direct solutions are found for this issue, as the NMR porosity did not match the porosity of many core-plugs within the core. One possible solution is to prioritize image-log resistivity cutoffs over the NMR porosity ones, but that might potentially overestimate good sands. Another solution is to acquire high-resolution porosity or permeability curves. However, no such technology exists to this day as both porosity and permeability are typically indirectly inferred through different well logs measurements.

Results

Many factors play a major role in clastic reservoir quality. These include depositional conditions, tectonic setting, provenance, and diagenesis. The first control on the reservoir quality under study is depositional. This conclusion was reached by comparing thin-sections, XRD, porosity and permeability, and MICP results of two distinct samples (Figure 7). The thin-section for the sample on left in Figure 7 shows a relatively well-sorted, very fine-grained matrix. XRD (Table 1) shows a dominance of calcite (72.7%), and lower percentage of quartz (~10%). Permeability of the sample is very low (below the tool's threshold to measure = 0.000 mD). As expected, MICP shows very narrow pore-throats that range between 0.01 and 0.001 microns, with the majority of the pore-throats at ~0.03 microns. The thin section for the sample on the right in Figure 7 shows poorly sorted, distinctly coarser grained matrix. XRD (Table 1) shows a lower percentage of calcite (25%) and higher percentages of quartz (46%). Permeability of the sample is low (0.010 mD), an order of magnitude higher than the fine-grained sample. MICP supports this measurement by showing wider pore-throats ranging between 0.005 and 1 microns, with the majority of the pore-throats at ~0.3 microns. The analysis done on these two samples shows that generally with increasing grain-size, pore-throats will be wider.

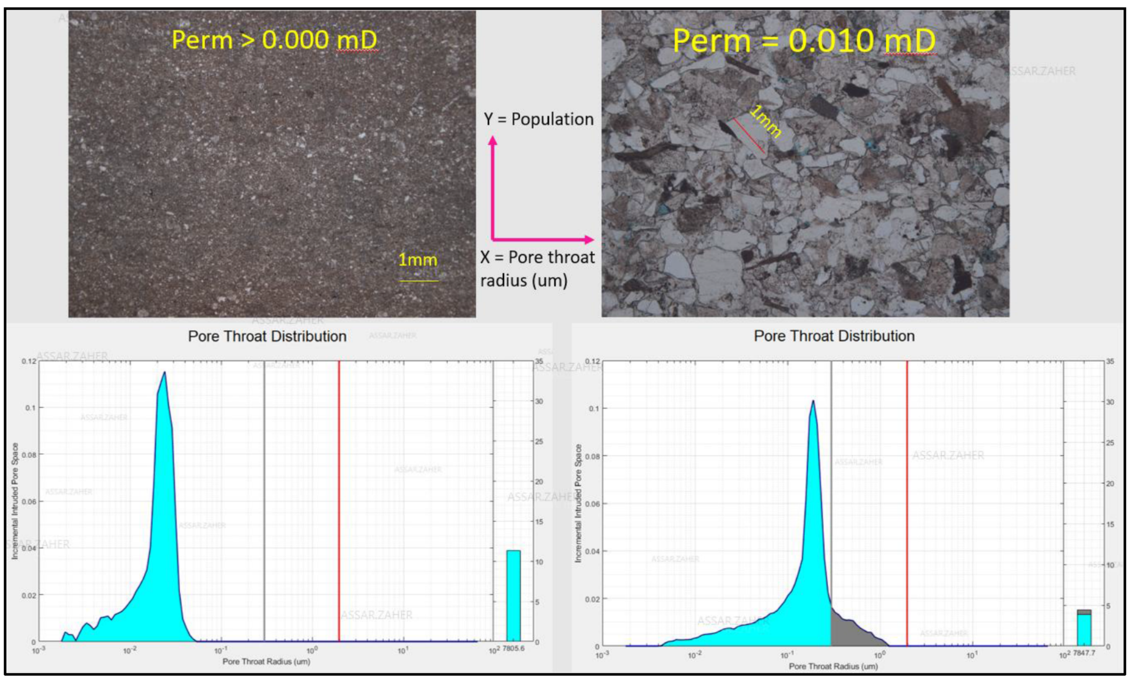

Figure 7—Diagenetic control, mainly cementation/dissolution, highly impacts pore-throat radius. Sample P1003 (left) shows very low permeability associated with heavy cementation. Sample P1002 (right) shows increasing permeability with some dissolution.

The second control on the reservoir quality under study is diagenetic. The thin-section of the sample on the left (Figure 8) shows a poorly sorted, low maturity, coarse-grained matrix that is heavily cemented. XRD (Table 1) shows a dominance of calcite (56%), with minor dolomite (6%) and a low amount of quartz (13%). Permeability of the sample was very low (below the tool's threshold to measure = 0.000 mD). MICP shows a pore-throat distribution that ranges from 0.01 to 0.5 microns. It also shows two different populations of pore-throats, one at ~0.02 microns and another at ~0.3 microns. The thin-section of the sample on the right (Figure 8) shows a poorly sorted medium to coarse-grained matrix, and partially dissolved cements. XRD shows 36% quartz, 8% calcite, 9% dolomite, 11% K-feldspar and 19% plagioclase. Permeability of this sample is 20 mD, at least 3 orders of magnitude higher than the other sample. The MICP shows a wide distribution of pore-throats, ranging from 0.01 to ~ 8 microns, with the majority of the pore-throat population at ~7 microns. The analysis done on these two samples, and others (Table 1) shows that generally with increasing grain-size and dissolution, pore-throat radius will increase.

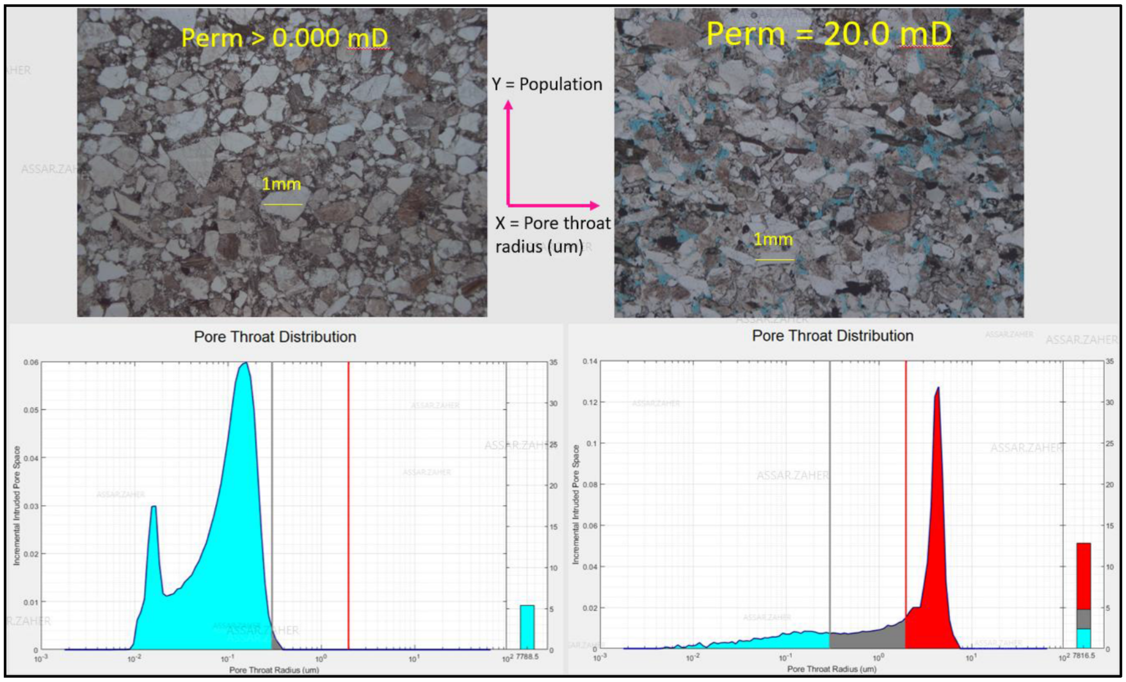

Figure 8—Diagenetic control, mainly cementation/dissolution, highly impacts pore-throat radius. Sample P1008 (left) shows very low permeability associated with heavy cementation. Sample P1006 (right) shows increasing permeability with some dissolution.

Table 01—Full XRD data on samples taken from 9 core-plugs.

Well	ID	Plagioclase	Anhydrite	Barite	Calcite	Chlorite	Dolomite	Halite	Hematite	Illite/mica	Kaolinite	K feldspar	Pyrite	Quartz	Siderite	Total
		%	%	%	%	%	%	%	%	%	%	%	%	%	%	%
Well-1	P1001	10.6	1.5	0.2	38.4	0.8	1	3	0.5	4.7	2.8	5.9	0.5	29.5	0.6	100
Well-1	P1002	8.8	1.6	0.2	55.7	1.9	6.1	1.7	0.1	2.8	2.4	3.8	0.7	13.2	1	100
Well-1	P1003	4.9	1.2	0.4	72.7	2	0.6	0.2	0.2	4.3	2.5	0.8	0.4	9.8	0	100
Well-1	P1004	3.7	1.1	0.2	59.3	1.5	4.1	0.6	0.5	5.7	2.2	4.4	0.5	16.2	0	100
Well-1	P1005	19.6	2.5	0.3	18.7	2.4	3.1	1.9	0.8	8.6	2.1	13.2	0.2	26.6	0	100
Well-1	P1006	18.8	4	0.2	8.1	2	8.9	0.6	0.9	6.8	2.2	10.8	0	35.7	1	100
Well-1	P1007	18.2	4.7	0.3	4.7	1.6	7.1	0.5	0.9	6.8	2.4	10.1	0.4	40.6	1.7	100
Well-1	P1008	11.8	2.8	0.2	25.2	0	2	0	0.2	4.2	2.5	4.1	0.7	46.3	0	100
Well-1	P1009	12.5	2.3	0.2	44.9	3.1	17.2	0.7	0.7	3.8	1.4	1.3	0.3	11.6	0	100

A rock-type model was generated through the steps explained earlier in the paper. The supervised rock-type model test matched 37/46 (80.4%) of the core plugs permeability measurements. The unsupervised rock-type model test matched 33/48 (68.75%) of the core plugs permeability measurements. The model was blind-tested against side-wall cores, and matched their permeability values (Figure 9). The model was also blind tested against 19 core plugs permeability measurements at Well-2, and matched 14/19 permeability measurements (73.7%). Since the model consists of five rock-types, the good-sand rock-type can be considered net-sand, and the four remaining rock-types are considered non-reservoir. For more optimistic NTG calculation, good-sand and shaly-sand rock-types can be considered net-sand, and remaining rock-types are non-reservoir. The rock-type model can now be used for more representative NTG calculations, which are then used for improved net rock volume estimations.

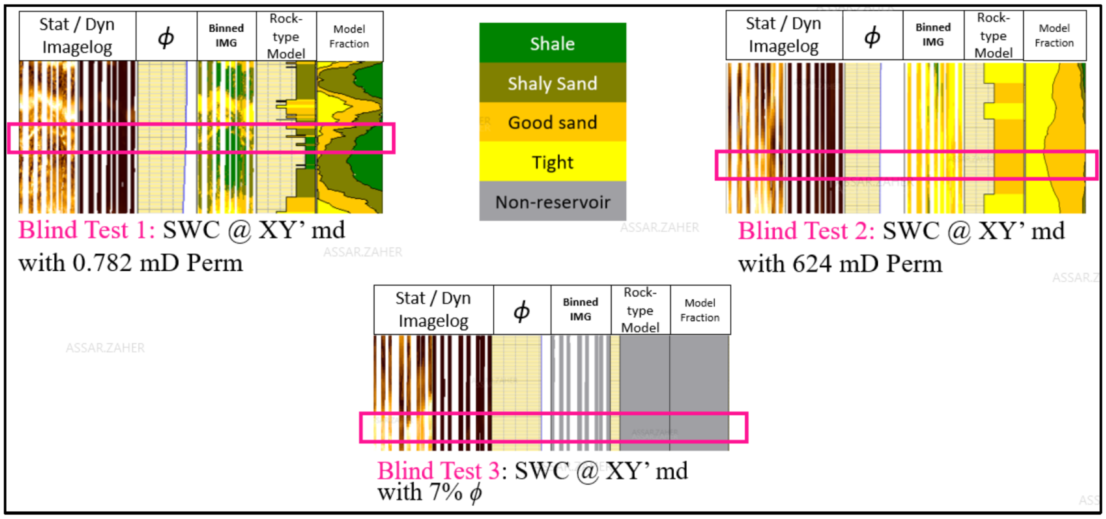

Figure 9—Blind-testing the rock-type model against random side-wall cores with P&P analysis conducted on them.

Ringrose & Bentley (2015) argue that models should be "fit-for-purpose". They promote the concept of building a direct model that addresses a specific question. The modeler should have a clear understanding of the problem and a model purpose that justifies their efforts. The model should be built by utilizing the proper tools for the problem. The hydrocarbon industry has taken a habit of following rigid workflows that do not necessarily include the proper tools or even address the underlying problems.

While Ringrose & Bentley (2015) target 3D reservoir models pitfalls, the same can be argued for petrophysical models. Conventional models that utilize low-resolution logging tools provide a proper answer in simple reservoirs, but fall short when evaluating heterogeneous thin-bedded reservoirs. The method proposed in this study attempts to design a model that fits the purpose of "counting" the amount of reservoir rock with enough porosity and permeability to produce hydrocarbon. Therefore, it can be argued that a model's success should be measured against its ability to count good sands. The model's failure to count specific non-reservoir rock-types should not matter, as it is not the purpose of the model. Under this rationale, the model should be validated against its ability to distinguish reservoir from non-reservoir rock-types. Reservoir rock permeability cutoff was defined as 1 mD as the lowest value that could flow hydrocarbons. This value was established as the cutoff for the wireline formation tester's (WFT) sampling success in the industry. The same logic is also applied when validating the ANNs predictions in Well-2 against core data. The rock-type model matched 80.4% of the calibration points from the core. This correlation is acceptable considering the limitations and uncertainties discussed in methodologies, but a higher correlation can be achieved. A proper acquisition plan that addresses the uncertainties of the method should lead to higher correlations and give more confidence in the model.

In light of the results, it is worth reviewing big data applications, including machine learning as it is becoming more applicable with the continuous advancements in computer technology. The quick turn-around time of results allows for thorough analysis in reasonable durations. Artificial neural networks were the main form of machine learning used in this project. However, ANNs do not come without risk and uncertainties. This is particularly true when considering that Well-2 is a few kilometers away from the location of Well-1. This fact highlights a few concerns for investigation and discussion. Three conditions were set for the machine to apply its learnings from Well-1 to Well-2: a) Well-2 should have the same set of logs as Well-1, b) Well-2 should have similar depositional conditions to Well-1, c) Well-2 should be in close proximity to Well-1, to increase the chances of it undergoing the same diagenesis conditions. These conditions and their potential impact on predictions will be addressed in the following discussion.

As previously mentioned, the supervised testing using ANNs correlated 88.7% to the reference rock-type model. The rock-type model in turn matched 37/46 (80.4%) of its calibration points. This means that the

ANNs are being referenced to a model that's close to the truth, but does not completely represent it. Adding to this inherent uncertainty, ANNs use logs of lower resolution than the ones utilized by the model. The problem that this thesis sets to solve involves thin-beds of heterogeneous reservoir quality. Some logs have the vertical resolution of 2'-4' and completely miss the high-resolution reservoir properties changes.

To compensate for this issue, both the reference and the SRES log are given enough weight in the ANNs input parameters to control the prediction. Lower resolution logs are given less weight, but not completely neglected. As mentioned before, the rock-type model is not a complete representation of the truth. Also, figure 4 shows that SRES captures thin-beds well, but is very noisy when it comes to low-resolution changed in homogeneous lithology. This is where the lower resolution logs come into play, as they resolve these thicker intervals better than SRES and in some locations even better than the rock-type model.

The supervised test's prediction at Well-1 matched 68.75% (33/48) of the rock-type model's calibration points. The result is expected because of the thin-bedded heterogeneity of the reservoir, and how low-resolution logs interact with it. With enough time and training, a higher correlation to the model or directly to the core's calibration points is sure to be made. Unfortunately, the allocated time in this study did not allow for full expansion on the correlation potential of ANNs. Instead, the prediction results presented in this thesis are of limited experimentation with indexation and weighting of input parameters.

Once ANNs reached a high enough correlation to the rock-type model in Well-1, their learnings were applied to Well-2. The prediction matched calibration points from cores in Well-2 around 73% of the time. The prediction is considerably good, since the evaluated section at Well-2 is of similar depositional environment to Well-1. The reservoir is thinly-bedded and highly heterogeneous. Some key logs in Well-2 were missing, most notably the NMR porosity which proved to be excellent for ANNs to learn from in Well-1. Missing NMR porosity in Well-2 should theoretically lower the correlation between prediction and core data. This assumption however cannot be verified without NMR data in the well. The work of He et al (2019) supports this assumption, as they had a full suite of input logs, their prediction's accuracy ranged between 66 to 99%. Conversely, the accuracy of their predictions diminished between 45 and 98% whenever their input data were missing key logs. Fortunately, SRES and image-log were present in both wells and without them the prediction would have most likely suffered in such a heterogeneous reservoir. Thorium concentration log was very noisy and caused many issues in the correlation. It was eventually removed from the input parameters to reach a higher correlation with both rock-type model in Well-1, and with core data in Well-2.

Ben-Awuah & Padmanabhan (2017) utilized ANNs to differentiate between the reservoir quality of 5 types of sands. The desired prediction of their work was to rock-type different sands based on their permeability. At first, they experimented with a combined facies prediction model. The model yielded a low correlation coefficient of 41% between predicted and measured permeability from core data. Instead of a combined model, they developed models based on single facies. By doing so, the predicted permeability reached an accuracy up to 99%. They had particular correlation issues with bioturbated sands, and the highest predicted permeability correlation they reached was 85%.

While the aim of Ben-Awuah & Padmanabhan's (2017) work is similar to this study, the tools they utilized and the reservoir conditions they worked with differ greatly. Most sand intervals they worked with were of thicker interval than the vertical resolution of the logging tools used (+1'). This gives confidence to low-resolution logging tools to predict without overlooking or averaging-out reservoir intervals. They also utilized three wells from three different fields for the ANNs to learn from, and one well to apply the learnings to. It is not discussed based on what criteria these wells were selected, apart from them having cores and core-associated data. On the contrary, this study only utilized one well for ANNs to learn from, and one to apply learnings to. This is due to the lack of well data in a frontier exploration basin. With the excellent correlation results from their work, it is possible that this study can achieve higher prediction correlations with more data from similar reservoirs.

Interestingly enough, the work of He et al (2019) contradicts the statement mentioned above. They used 8 wells to conduct their study, and correlations between prediction and measurement ranged between 66 to 99% at best. The study dealt with a total core thickness of ~2000m (6561.68’) and 10 different facies for ANNs to predict in a combined model. With such a large amount of facies for ANNs to predict, and only 37.2% of the wells having full input logs, it becomes understandable as to why more data doesn’t produce better ANNs predictions. Apart from unavailability of logs, the process of acquiring log and correcting them, interpreting core facies, and producing core-plug measurements introduce a lot of uncertainty that is difficult to quantify and incorporate into the predictions. If these factors are addressed early in an ANNs project, higher prediction correlations will be achieved. The exploration campaign in the area of this study should also design well logs acquisition to fulfill the requirements of ANNs analysis.

As mentioned previously, one of the conditions for ANNs prediction is a similarity in depositional conditions. The word "conditions" is used instead of environments because different environments can deposit similar reservoirs. Passey (2006) pointed to tidal-flat and turbidite formations depositing thin-bedded reservoirs. Other depositional environments also yield thin-bedded reservoirs like meandering rivers (Jordan & Pryor, 1992). If provenance and depositional processes of each environment are similar, it is reasonable to assume that logging tools will react similarly. This basic assumption is behind the logic of using ANNs as tools for prediction across wells. Well-2 and Well-1 are both interpreted to be of similar depositional environments, and the prediction that matched 73% of core data in Well-2 supports the assumption.

The similarity of depositional conditions is particularly important in this study because only two wells were used. Well-1 to learn from, and Well-2 to apply learnings on. Many prediction problems would occur if Well-2 is of different depositional conditions. ANNs would be requested to predict facies that they do not recognize. Similar log signatures could indicate different reservoir properties depending on the type of reservoir and fluids that are stored within it.

There are over 40 minerals that could act as cementation agents in sandstone (Ali et al., 2010). The most prevalent of which are calcite, quartz, anhydrite, dolomite, hematite, feldspar, siderite, gypsum, clay minerals and barite (Ali et al., 2010). Calcite is typically deposited in shallow water as a result of living organisms precipitating carbonates (Ali et al., 2010; Morad et al., 2010). Deep-marine turbidite deposits are the result of reworked shallow-marine materials that may contain calcite within them (Morad et al., 2010). Calcite can exist in sandstones as carbonate bioclasts, or dissolved within the high-salinity pore water (Morad et al., 2010). Calcite’s presence in deep-marine turbidites causes extensive cementation of sandstone reservoirs (Morad et al., 2010). This cementation effect can be observed in the XRD values of table 1, where calcite presence ranged between 4 to 72%. It can also be observed in thin-sections at samples #1002 and #1008 (Figures 7 and 8).

It was established that ANNs cannot predict rock-types they did not originally have in their training set of data. To reduce the uncertainty of having different rock-types between wells, it is assumed that wells should be relatively close to each other. If Well-1 and Well-2 both included reworked shallow-marine sand intervals, they should undergo similar diagenesis processes. Diagenesis is controlled by many factors, including: 1) original depositional facies and associated fluids composition, 2) sandstone composition, 3) rate of sedimentation, 4) burial and thermal history of the basin (Morad et al., 2010). While most of these factors are beyond the scope of this study, the argument that closer wellbores might undergo similar diagenesis can be made.

Exceptions to this claim can happen for various reasons. For example, turbidites in Well-2 could be deposited much deeper than Well-1. This will introduce differences in compaction, temperature, and even reservoir maturity. These factors in turn will affect how logs acquire their readings in these intervals, and might even produce rock-types that are not present in Well-1. Another example is tectonic movements, provenance, and temperature and therefore reservoir composition and maturity will be different if Well-2 is located in a different structural block than Well-1.

In this project, Well-2 is located ~4 km away from Well-1, and the reservoir is ~2000' deeper in Well-2 than Well-1. The core porosity and permeability analysis done on Well-2 show a lot more cementation and much lower permeability values. This could be a factor of increasing depth and temperature to the same provenance material, or completely different provenance that is of lower reservoir quality and more proneness to cementation. Sedimentological models in the area support the first assumption. Fortunately, tight and cemented sands were present in some intervals of Well-1. Their presence allowed the ANNs to recognize them in Well-2 and achieve a reasonable correlation between core-data and prediction.

Conclusions and Recommendations

Low-resolution logging tools do not completely resolve thin-bedded reservoirs for calculating net reservoir. To mitigate these issues the study proposed a three-fold approach: 1) MICP and core-data analysis to investigate underlying causes for thin-bedded reservoir heterogeneity; 2) integrating core and image-log data to produce a representative petrophysical model; and 3) investigating the feasibility of employing machine learning models to both nearby and future wells.

MICP analysis of core plugs showed that the heterogeneous reservoir quality improves primarily with increasing grain size, and secondarily with dissolution. Similar analysis should be performed on other reservoirs to investigate different controls on heterogeneity. The core-calibrated petrophysical model predicted good reservoir quality rock-types with an accuracy of 80.4%.

Further work is required to increase the accuracy of future rock-type models and artificial neural network (ANN) predictions. This includes:

- Studying the diagenetic history in the basin to estimate the lateral extent of certain diagenetic processes that affect reservoir quality.
- Quality checking the laboratory measurements on core-plugs is essential for reducing uncertainty in MICP analysis and ANN predictions.
- Acquiring the proper set of data for ANNs in future wells potentially allowing better predictions.

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